



The role of the metal-oxide Interface's terminating layer on the selective cold cracking of a commercial Niobium–Hafnium–Titanium (C-103) alloy

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ARTICLE INFO

Article history:

Received 12 July 2020

Received in revised form

28 September 2020

Accepted 30 September 2020

Available online 1 October 2020

Keywords:

Niobium

Cold cracking

Interface

EBSD

DFT

ABSTRACT

A commercial Niobium–Hafnium–Titanium (Nb–Hf–Ti) alloy (C-103), produced by vacuum arc remelting (VAR), showed clear signatures of internal oxidation viz. presence of monoclinic-HfO₂ at all the grain boundaries. Minor cold rolling, to ~10% reduction in thickness, initiated cold cracking. This was observed at some, but not all, of the high angle Nb–Nb grain boundaries containing coarse HfO₂ 2nd phase oxide precipitates. Closer microscopic examination revealed cold cracking at selective Nb–HfO₂ interfaces. There appeared no clear mesoscopic experimental rationale behind this selective interfacial cold cracking. In order to understand this, density functional theory (DFT) calculations were performed on the metal-oxide interface models. They showed significant separation or decohesion energy difference (~76%) with different atomistic bonding at the terminating layer of the Nb–HfO₂ interface. In particular, the interface with one atomic layer of Nb bonded with oxygen of HfO₂ had very low work of separation, which was about ~8% lower than the Nb–Nb interface of the bulk Nb. In brief, the DFT simulations brought out the defining role of metal-oxide interface's terminating layer on the interface decohesion. This is in agreement with the past theoretical research performed on several metal|non-metal interfaces. The present study extends this hypothesis to a practical example of selective cold cracking of a commercial Nb–Hf–Ti alloy.

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1. Introduction

Niobium (Nb) alloys retain reasonable strength, beyond the capabilities of Ni-base superalloys [1], at very high temperatures. Naturally, they find niche applications [2] in aerospace. Often the commercial Nb-alloys contain ~10 wt % hafnium (Hf) which provides excellent solid solution strengthening [3]; strengthening extending to temperatures as high as ~1400 °C [4,5]. Hf also acts as a sacrificial element for any internal oxidation [6,7]. Small quantity of Ti with other elements like Zr (in very small amount), on the other hand, improves the overall oxidation resistance [8,9]. This study involved internal oxidation and associated cold cracking at selective metal-oxide interfaces in a commercial Nb–Hf–Ti alloy (C-103). The cold-cracking was observed in the form of room

temperature cleavage fracture without significant plastic deformation.

One of the critical issues in a commercial Nb-alloy is its oxygen content [1]. Enormous care is taken during the processing to avoid internal oxidation [10]. The latter may appear as grain boundary (GB) HfO₂ 2nd phase oxide precipitates, and for higher oxygen content even as grain interior Widmanstätten oxides [11]. Past studies [12,13] have shown that oxides affect both strength and ductility. In complex forming environment, they appear to initiate cold cracking [14]. However, very little has been documented on the exact nature of the oxides and on the metal-oxide interface: especially on the possible role of such interfaces on cold-cracking in Nb, which is precisely the motivation behind the present study.

The starting material, a commercial C-103 alloy, had oxides (HfO₂) at all of its grain boundaries. Cold-cracking was observed in some (but not all) of the metal-oxide interfaces. Extensive micro-textural characterization failed to provide an apparent rationale behind the mechanism of selective cold cracking. In order to look

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into the details of the selective cold cracking, density functional theory (DFT) calculations were performed to explore the role of metal-oxide interface's terminating layer on the selective cold cracking of the alloy.

The nature and decohesion of the metals||non-metals (oxides/carbides/nitrides) (here '||' means interface) were determined by the atomic bonding of the terminating layer at the interface [15,16]. It has been noted that the electronic interactions, applicable at metal-oxide interface, involve strongly hybridized metallic, ionic or covalent bonding; all of which can be accounted for by the DFT calculations. In the past, *ab-initio* simulations have been used to understand the phase separation/decohesion between Nb||Al₂O₃ [17], Al||WC [18], Al||VC [19], Al||VN [19], Al||TiN [20], Fe||WC [21], γ -Ni(Al)|| α -Al₂O₃ [22], SiC||W [23] (see Table S1) and many other similar interfaces. Using the first-principles calculations, it was thus possible to calculate the work of separation (W_{sep}), representing the amount of reversible work per unit area of interface required to separate the two crystals by mechanical relaxation of atomic positions after separation and not allowing any diffusion. The W_{sep} of one particular interface viz. The ... NbNb||NbOHfO ... was found to be significantly lower than the ... NbNb||NbNb ..., ...NbNbOHf||OHf ... and ... NbNb||OHfO ... interfaces, which may provide a possible explanation for the experimentally observed selective cold cracking.

2. Experimental methods

A commercial C-103 alloy, chemical composition listed in Table 1, was produced by vacuum arc remelting (VAR) followed by light hot forging. A batch of production ended up with higher oxygen content: ~360 ppm (parts per million by weight) oxygen (O) when the maximum allowed oxygen concentration was only 100 ppm. Other impurities (example: carbon (C), nitrogen (N) and hydrogen (H)) were measured and found to be less than 0.01 wt% each. These trace level impurities are not expected to affect the mechanical properties [10]. Further, all the Nb–Nb grain boundaries showed signature of internal oxidation.

Hot forged alloy was cold rolled to various thickness reductions. This study, however, reported (for brevity) data on ~10% cold rolled specimen, where observations on cold-cracking and grain boundary HfO₂ were made with appropriate field emission gun (FEG) scanning electron microscope (SEM). Metallographic sample preparations were performed by polishing with sub-micron colloidal silica. These were then subjected to microscopic characterization involving a FEI™ Quanta 3 d-FEG (Q3d-FEG: for analytical microscopy) and a FEI™ Nova-Nano-SEM (NNS: for imaging). In particular, the Q3d-FEG was used for energy dispersive X-ray spectroscopy (EDS) and electron backscattered diffraction (EBSD) – EDAX™ systems. The NNS, on the other hand, provided imaging, especially imaging with CBS™.

The HfO₂ precipitates at the grain boundaries were extracted using dissolution of the test specimen in an HF and HNO₃ (1:1) solution. The residue was separated in a high-speed centrifuge, followed by filtration and then X-ray diffraction: a Bruker™ D8-Discover with micro-focus X-ray plus Vantec™ detector. Area detector signals from multiple 2nd phase clusters were obtained and collated, for further details the reader may refer to Refs. [32]. The

Table 1

Chemical composition (in weight percentage alloying element) of the C-103 alloy used in this study.

Nb	Hf	Ti	O
Balance	7.89%	1.38%	0.036% (360 ppm)

phase identification and quantification were performed through a commercial software Highscore-plus™.

3. Computational details

DFT calculations were performed using the projector augmented wave (PAW) method [24,25] as implemented in the QUANTUM ESPRESSO package [26], which is a plane wave based code. For the exchange-correlation functional the Perdew–Burke–Ernzerhof (PBE) [27] variant of generalized gradient approximations (GGA) was employed. Total energy of the systems was calculated by complete relaxation (geometric and atomic). The relaxation was performed using force convergence to less than 1×10^{-4} Ry/a. u. for all the structures. Monkhorst-Pack scheme with converged *k*-point mesh of $3 \times 3 \times 1$ was used in the entire supercell. Converged cut-off energy of 50 Ry was used for convergence of the total energy of the system. Davidson type diagonalization method was implemented with the convergence energy threshold of 1×10^{-9} Ry.

Since a significant proportion of the major alloying element (Hf) was internally oxidized and the region near the GB became deficient in Hf (as shown latter with analytical microscopy), the DFT calculations were limitedly performed on interfaces containing Nb and HfO₂, where also the cold cracking was observed. The supercells in the DFT simulations incorporated experimental Orientation Relationship (OR) between Nb and HfO₂ (2nd phase oxide precipitates). The experimental OR was obtained from the EBSD (also shown latter) as: (001)_{Nb}/(001)_{Oxide} and <001>_{Nb}/<103>_{Oxide} (here '/' means parallel to). The relaxed lattice parameter of the bulk Nb (*Im-3m*) and HfO₂ (*P2₁/c*), as listed in Table S2, were used as an initial configuration. In the plane of the interface, a 2×2 unit cell of HfO₂ ($a = 5.1034$ Å, $b = 5.1508$ Å) was placed on top of a 3×3 unit cell of Nb ($a = 3.304$ Å) in order to maintain the lattice mismatch to less than 3% (i.e. for Nb, 3×3.304 Å = ~9.9 Å and for HfO₂, 2×5.103 = ~10.0 Å) (see Fig. S1). The lattice parameters were found to be in agreement with the literature [35–38] (see Table S2).

Finally, W_{sep} , which represents the work that is needed for the breaking of the interfacial bonds and creation of two free surfaces, was calculated. Of course, this calculation does not account for the interplay of complex phenomena such as diffusion or plastic deformation. Eqn. (1) provides the formulation of W_{sep} [17–23] as,

$$W_{sep} = \frac{1}{A} (E_1^{slab} + E_2^{slab} - E_{12}^{tot}) \quad (1)$$

Where E_{12}^{tot} gives the total energy of the heterostructure comprising of Nb and HfO₂ multi-layered slabs. E_i^{slab} , $i = 1, 2$ gives the total energy of the slab containing one of the pure materials separated by vacuum of 10 Å, which is generally sufficient to ensure the negligible interaction between the periodic images and A is the interface area. For example, W_{sep} of Nb–HfO₂ (... NbNb||NbOHfO ...) interface was calculated by considering E_1^{slab} and E_2^{slab} as the total energy of the supercells with (... NbNb||vacuum) and with (vacuum||NbOHfO ...), respectively (as shown latter with schematic). Layer by layer convergence of W_{sep} for the interface ... NbNb||NbOHfO ... was checked starting from 2 layers in each slab (see Fig. S2 and Fig. S3). Following this, 7 layers of Nb and 10 layers of HfO₂ was chosen to build the heterostructures.

$$E_{surface} = \frac{1}{2A} (E_{slab} - NE_{bulk}) \quad (2)$$

To benchmark the DFT results, surface energy of bulk Nb (001) was calculated by eqn. (2), where, E_{bulk} is the total energy of the primitive unit cell, containing one atom per formula unit and E_{slab} is

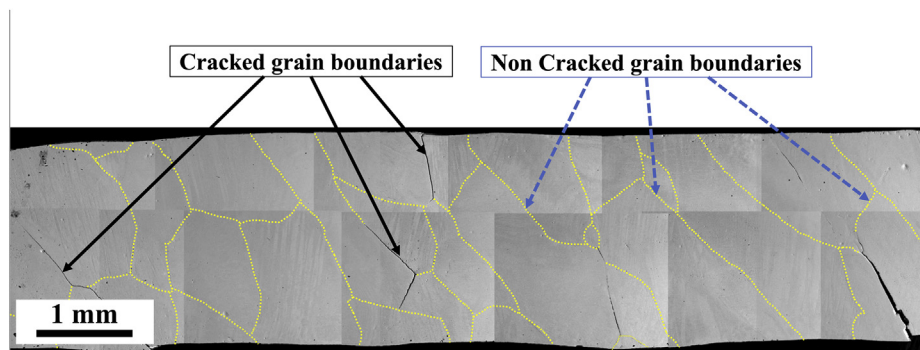


Fig. 1. SEM secondary electron (SE) image of the 10% cold rolled specimen: showing Nb–Nb grain boundaries with and without cold cracking.

the total energy of the *N*-atom 3D slab with 10 Å vacuum. *A* is the surface area of the slab exposed to vacuum. The surface energy obtained was 2.637 J/m², which is in good agreement with the previously calculated value [28] and the experimentally measured value of 2.6 J/m² [29].

4. Results and discussions

Fig. 1 represents a composite SEM micrograph of the 10% cold rolled C-103 sheet. Superimposed on the same micrograph, of visible cold cracking, are the Nb–Nb grain boundaries. It is to be noted that cracks were observed in ~20%, but not all, of such grain boundaries. It may be further noted that the cold cracking remained selective even at higher rolling reductions: a maximum of ~30% of the grain boundaries showing visible cold cracking even towards the latter stages (see Fig. S4). All the grain boundaries contained coarse precipitates, identified as hafnium oxide (HfO₂). However, a few (and not all) such precipitate-containing grain boundaries showed visible cold cracking. One such grain boundary containing HfO₂ precipitates, which was fractured along the grain boundary surface, is shown in Fig. 2. Cold cracking was observed only in the rolled specimens. In particular, this was restricted only to selective metal-oxide interfaces (see Fig. 3a). In particular, only ~20–30% (and not all) of the metal-oxide interfaces showed this cold-cracking (see Fig. 3b).

EBSD measurements were made on the hot forged structures, before and after the cold rolling. They showed presence of grain interior deformation banding (or narrow bands of strain localizations, see Fig. 4a). This, clearly an integral part of the cold deformed microstructures, was observed in grain clusters with and without

the cold cracking. Near-boundary regions without visible cracks showed (see Fig. 4b) misorientation developments or signatures of plasticity. Boundaries with cold-cracking, on the other hand, had insignificant misorientation evolution and hence limited plasticity. There was no clear distinction in the nature (axis-angle values) of grain boundaries with and without cold-cracking. Arguably, grain boundaries with cracks appeared to have a higher probability of possessing coarse 2nd phase (see Fig. 5a). However, many of the boundaries with coarse 2nd phase also did not show cold cracking. Fig. 5b shows that the GB misorientation angles are higher for boundaries with cracks than without cracks. In general, cold cracking was observed on some (and not all) of the Nb–Nb high angle grain boundaries containing coarse 2nd phase (HfO₂).

Non-aggressive dissolution of the matrix phase (as described in the experimental method) was used to extract the 2nd phase (Fig. 6). XRD data were obtained from many of such extracted particles, using micro-focus XRD and an area detector. Analysis of the XRD showed presence of majority (~90% by volume) monoclinic-HfO₂, and minority Nb plus hafnium-niobium oxide (Hf₆Nb₂O₁₇) phases. Analytical microscopy (see Fig. 7), by EDS, also confirmed the presence of monoclinic-HfO₂ at all the GBs. Finally, EBSD provided (Fig. 8a) crystallography of the GB HfO₂. It appeared that the oxide phase nucleated at the GB and then had grown. Clear growth twins decorated the oxide phase (also shown in Fig. 8a with point-to-origin misorientation distribution over the twinning region of the precipitate). This nucleation causes Hf deficient regions adjacent to the GBs (shown in Fig. 7). The oxide phase typically had an OR with the matrix Nb. This is shown in Fig. 8b as (001)_{Nb}//(001)_{Oxide} and <001>_{Nb}//<103>_{Oxide}. The experimental crystal structures and OR were used for the subsequent DFT calculations.

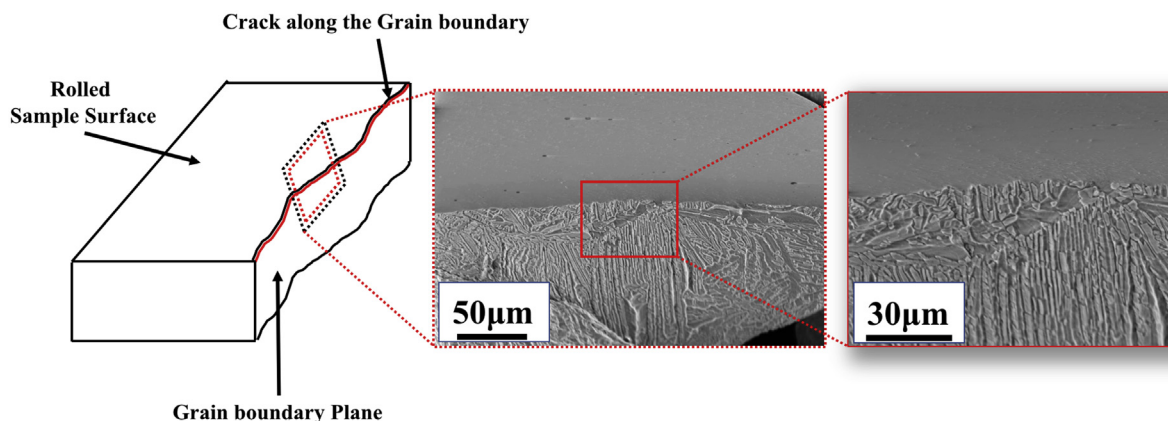


Fig. 2. SEM secondary electron (SE) image of precipitates decorating the Nb–Nb grain boundary plane of 10% cold rolled specimen.

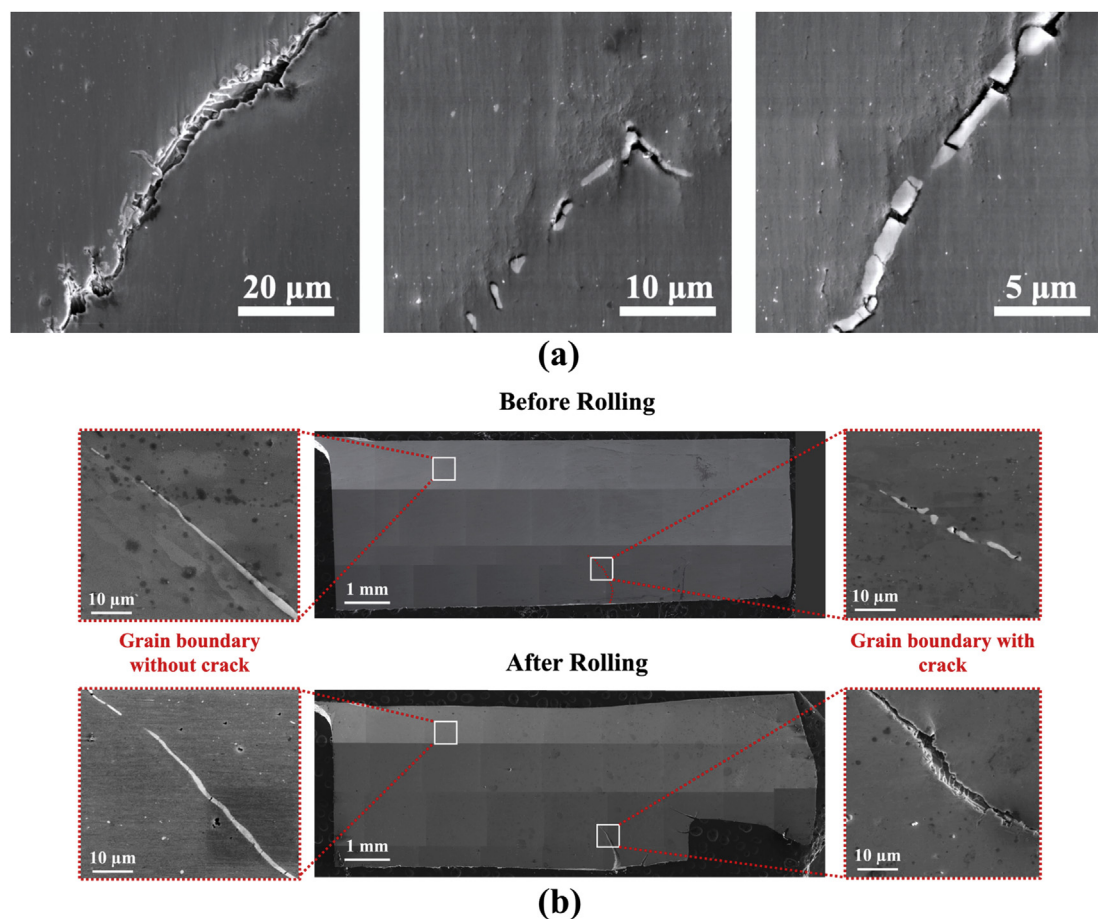


Fig. 3. (a) SEM backscattered electron (BSE) images highlighting grain boundary 2nd phase (latter identified as HfO_2) on a 10% deformed specimen. They decorated all boundaries – with and without visible cracks. (b) SEM images showing separation/decohesion of the selective (and not all) two phase (metal-2nd phase) boundaries after plastic deformation.

Though experimental mesoscopic EBSD data (from the present study) were extensive, they failed to explain the mechanism of crack initiation. Analytical microscopy, with EDS, also did not indicate any difference in the interface chemistry. It was hence inferred that along with the precipitate (ppt) size, and misorientation relationship, difference may exist in the atomistic nature of the interfacial chemistry and bonding. Such an aspect would be extremely difficult to establish experimentally, and *ab-initio* atomistic simulations were used for gathering further insights into the experimental observations.

The Nb- HfO_2 interface, with appropriate OR: $(001)_{\text{Nb}}// (001)_{\text{Oxide}}$ and $\langle 001 \rangle_{\text{Nb}}// \langle 103 \rangle_{\text{Oxide}}$ (Fig. 8b), was modelled for the DFT calculations. Fig. 9 shows the schematic of the Nb- HfO_2 interface. The difference in the nature, and hence energy, of such an interface would depend on the bonding of the terminating layer [17–23]. The W_{sep} was calculated for five different terminating layers and for bulk Nb (see Table 2) using eqn. (1). The interface ... NbNbOHf|OHf ... was found to show the highest W_{sep} value (9.63 J/m^2). This implied that a cleavage in the HfO_2 crystals was hard to occur due to strong covalent bonding between Hf and O atoms. Whereas, in case of the termination of ... NbNb|NbOHfO ..., the W_{sep} was found to be the lowest. The difference in the W_{sep} values between these two interfaces was found to be ~76%. The interface preference of Nb surface formation after the cleavage of ... NbNb|NbOHfO ..., can be rationalized from this difference in the W_{sep} . The difference between the W_{sep} of ... NbNb|NbOHfO ... (4.33 J/m^2) and that of bulk ... NbNb|NbNb ... (4.7 J/m^2) is not significant, but was still decisive

for the cleavage. In case of Nb–Nb, the bonding is metallic whereas, in case of Nb–O, it is ionic because of transfer of electrons from Nb to O, which justifies the higher W_{sep} in the latter case. Thus, the grain boundaries where HfO_2 is formed with O interface with the Nb metal, was found to be more susceptible to separation or cold cracking on deformation. However, all the other interfaces were found to show significantly high (at least greater than ~8%) W_{sep} as compared to ... NbNb|NbOHfO ... interface. Thus, they were not prone to separation. Although experimentally, it appeared that the separation had taken place at the interface between Nb and O terminated HfO_2 (... NbNb|OHfO ...), the DFT computed W_{sep} values revealed the termination to have taken place at the bulk Nb at Nb terminated HfO_2 interface (... NbNb|NbOHfO ...).

In order to understand the interaction at Nb- HfO_2 interface in more details, Bader charge analysis was employed [30,31]. The net average Bader charge on the atoms of different layers is compared in Fig. 10. The net average positive charge on Nb1 atoms (bonded to the O1 atoms of the interface) was found to be much higher than that of Nb atoms in the subsequent layers away from the interface (i.e. Nb2, Nb3). Thus, the Bader analysis confirmed that a net charge transfer has taken place from the Nb1 to O1 (whose net average negative charge content is higher than the O2 and O3 atoms). This confirms the bonding between Nb1 and O1. Thus, it implies that the most relaxed cleavage out of all possible termination was one with Nb saturated surface i.e. the separation had taken place by leaving one atomic layer of Nb attached to HfO_2 (... NbNb|NbOHfO ...) at the interface.

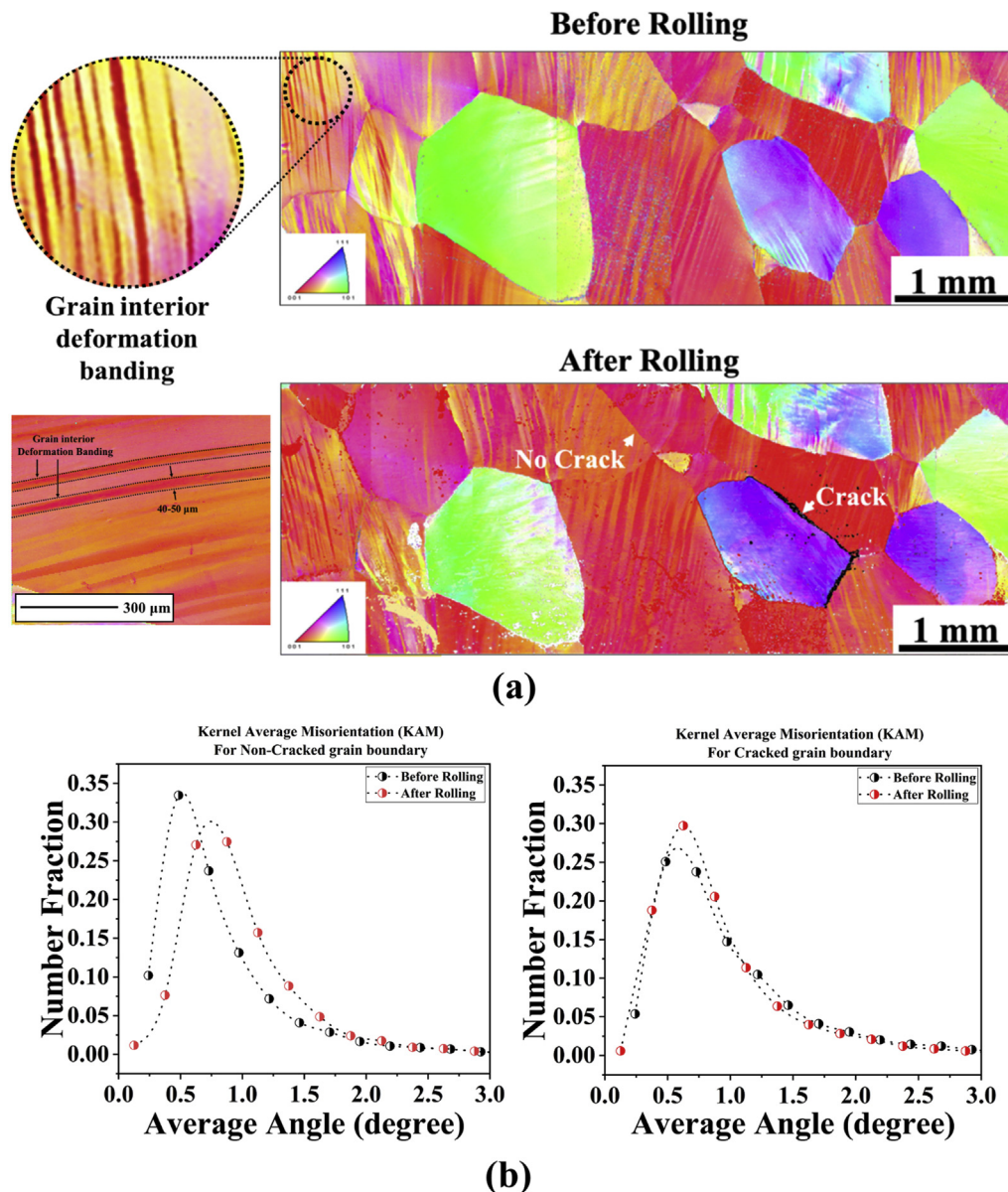


Fig. 4. (a) EBSD inverse pole figure (IPF) maps showing grain structures and boundaries before and after cold rolling. Grain interior deformation banding is also highlighted. (b) Kernel average misorientation (KAM) plots of identical near boundary regions before and after cold rolling. They show noticeable shift in KAM for boundaries without crack, while KAM distribution for boundaries with crack remained similar.

The present study thus acts as an interesting practical extension to the exhaustive past research [17–23] on decohesion of metal||non-metal (oxide/carbide/nitride) interfaces. The presence of HfO_2 precipitates definitely played an important role in the localized plastic deformation. The selective (10–20% of the total) interfaces of Nb– HfO_2 were subjected to decohesion without any plastic deformation. Arguably, these were coarse 2nd phase particles existing at high angle Nb–Nb grain boundaries. However, decohesion was not observed in many such interfaces. The analysis of interface separation energies based on W_{sep} , as shown in Table 2, thus provided a plausible explanation for the interesting experimental observation on selective cold cracking.

It is generally believed that the presence of coarse oxide/carbide/nitride particles is detrimental to the ductility of metallic materials [32–34]. This has been attributed to the formation of

particle deformed zone. Prakash et al., for example, showed that prolonged heat-treatment, which enabled partial dissolution of the 2nd phase, improved the ductility significantly, without sacrificing the strength [32]. This was attributed to a ‘scaling’ of dislocation mobility and a reduced severity of a particle deformed zone. Another possibility, emerging out of the present study, which remains scientific and not technological at this stage, is to tailor the interface chemistry. Ashby [34] in his classical 1966 paper, while proposing a model for dispersion-hardened single crystals, recognized the role of interface for relative plasticity and failure or decohesion. The present study, which is a continuation of past exhaustive research [17–23] on theoretical simulations, brought out the possibility of extending DFT simulations in exploring engineering failures.

Of course, measurements like 3 d atomic probe or HR-TEM

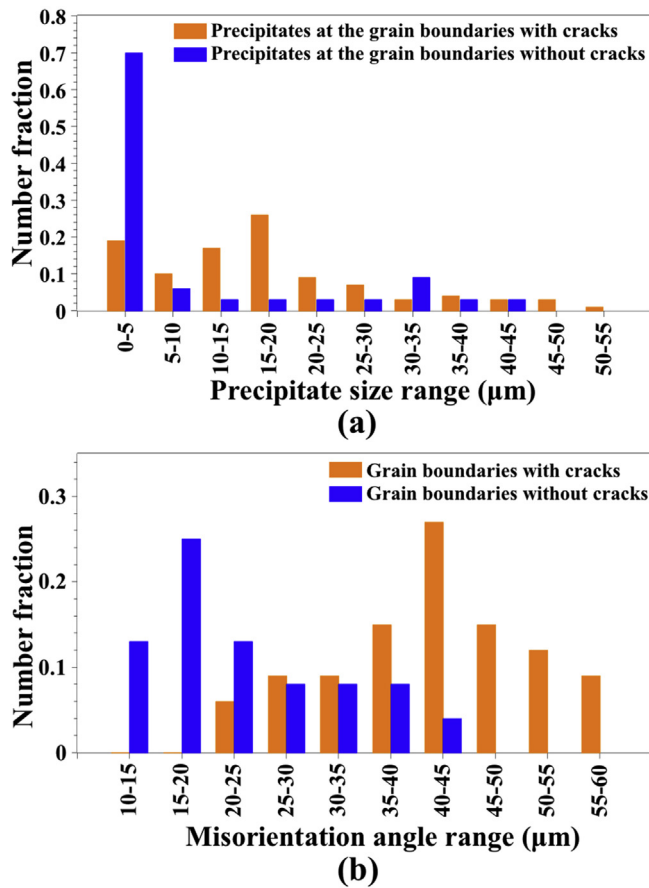


Fig. 5. (a) Precipitate size distributions, and (b) grain boundary misorientation angle distributions for Nb–Nb boundaries with and without cracks.

(transmission microscopy) might offer experimental possibilities. However, any such experimental measurements are 'limited' in scale and by the area of exploration, and it remains debatable if they can be successfully utilized to identify the role of terminating layer in the metal-oxide interface bonding. DFT calculations (Table 2 and Fig. 10), on the other hand, provided a potential

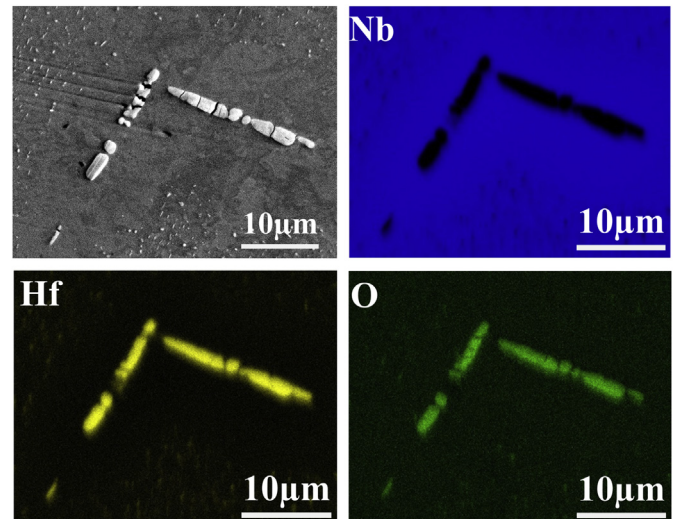


Fig. 7. Elemental mapping (with EDS) confirming presence of grain boundary HfO_2 highlighting the Hf deficient region.

explanation to the experimentally observed selective cold cracking of the metal-oxide interface. In particular, they brought out the defining role of the metal-oxide interface's terminating layer on the W_{sep} or interface decohesion.

5. Conclusions

A commercial Niobium–Hafnium–Titanium (Nb–Hf–Ti) alloy (C-103) was produced by VAR. The alloy had 360 ppm oxygen, and showed internal oxidation at all GBs. The GB oxide phase was identified as monoclinic- HfO_2 , and an OR $(001)_{\text{Nb}} // (001)_{\text{Oxide}}$ and $\langle 001 \rangle_{\text{Nb}} // \langle 103 \rangle_{\text{Oxide}}$ was experimentally established. Subsequent cold rolling led to cold-cracking, where ~20% (but not all) of the metal-oxide interfaces showed visible cracking or decohesion without any plasticity. The only difference found in the GB with and without cracks were the ppt size and the misorientation between the adjacent grains. Although this gives us the partial understanding of selective cold cracking, they failed to explain the mechanism by which it occurs. DFT calculations were employed to

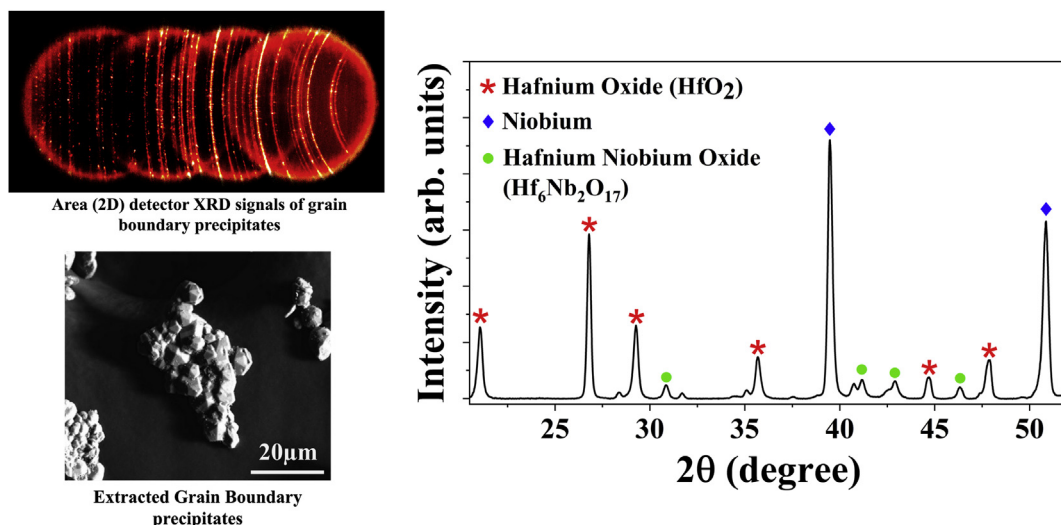


Fig. 6. XRD with area (2D) detector signal for the extracted 2nd phase. Phase identification shows presence of majority HfO_2 ($P2_1/c$) and minority $\text{Hf}_6\text{Nb}_2\text{O}_{17}$ ($Pbcn$).

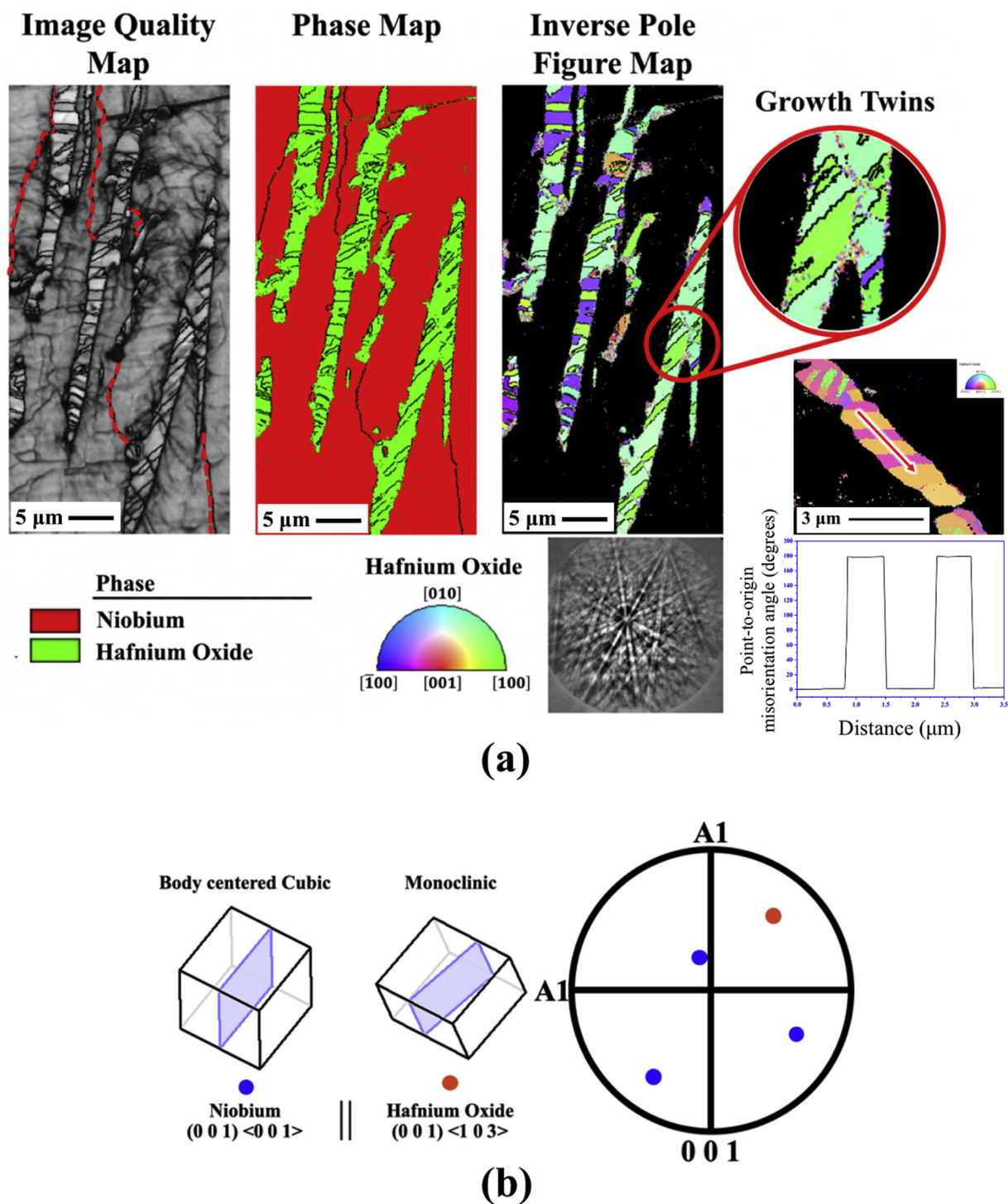


Fig. 8. (a) EBSD of grain boundary HfO_2 : showing the image quality (IQ), phase and IPF maps. Also included is a Kikuchi pattern of the grain boundary phase. Highlighted region showing growth twin. (b) Experimentally (from EBSD) obtained orientation relationship between bcc-Nb and monoclinic- HfO_2 .

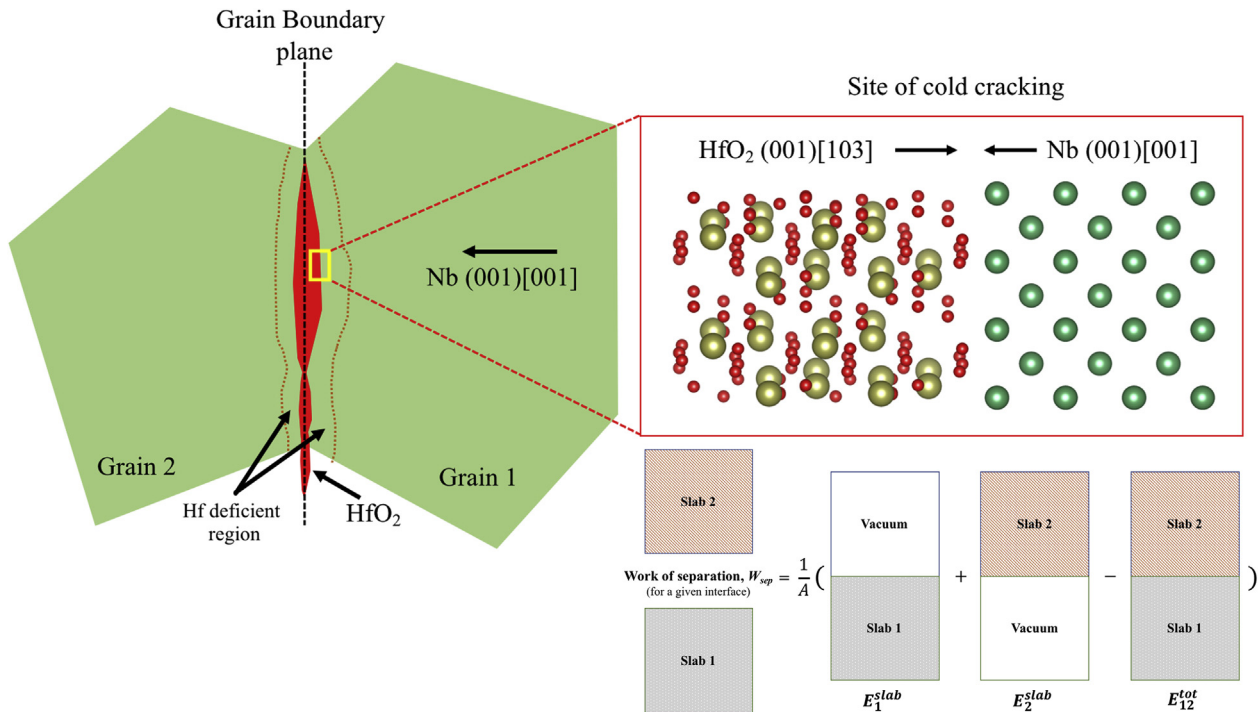


Fig. 9. Representation of a typical grain boundary with grain boundary precipitate and highlighted the Nb (100) and HfO₂ (001) interface based on the experimental orientation relationship (OR) taken for DFT calculations. Schematic representation of the simulation performed for calculating the W_{sep} of a given interface.

Table 2
Work of separation (W_{sep}) at different terminating interfaces.

Interface	W_{sep} (J/m ²)
... NbNb NbNb ...	4.7
... NbNbOHf OHf ...	9.63
... NbNb OHfO ...	7.03
... NbNb HfOHf ...	6.98
... NbNbO HfOHf ...	6.3
... NbNb NbOHfO ...	4.33

explore a potential explanation. It was shown that the chemistry of the terminating layer significantly affected the work of separation (W_{sep}) and hence decohesion. More specifically, it established that interface with one Nb atomic layer bonded with oxygen (*which might not be feasible to capture experimentally*) is weak and provides a site for easy separation. The study thus established the defining role of metal-oxide interface's terminating layer on the interface separation/decohesion, in general, and on cold-cracking of commercial Nb-alloys, in particular.

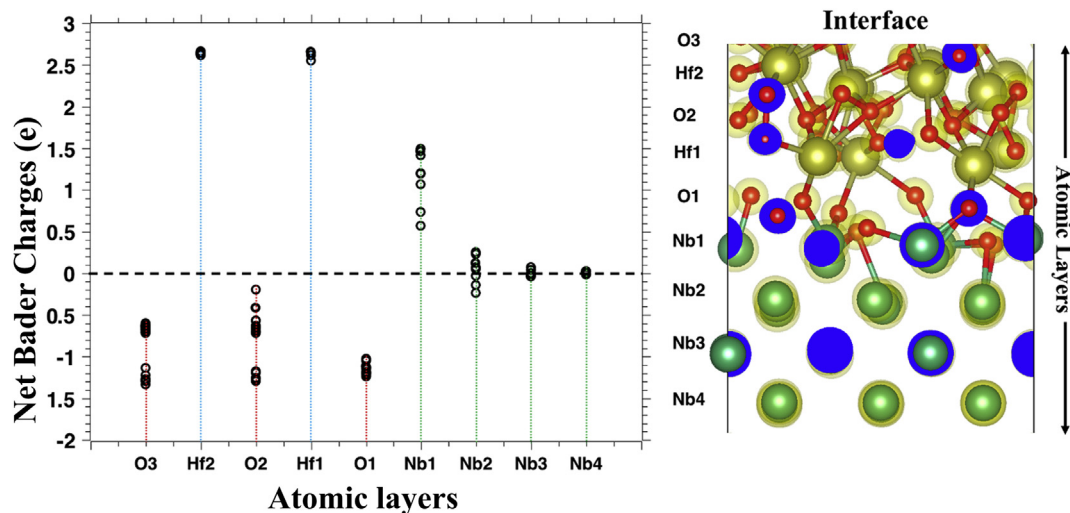


Fig. 10. Net Bader charges on Nb atoms of each layer is shown with electronic charge distribution maps.

CRediT authorship contribution statement

Ashish Dhole: Data curation, Methodology, Software, Formal analysis, Writing - original draft, Visualization, Methodology. **Amrita Bhattacharya:** Conceptualization, Software, Validation, Writing - review & editing. **Rohit Kumar Gupta:** Project administration. **Amol A. Gokhale:** Conceptualization, Methodology, Funding acquisition, Supervision, Writing - review & editing. **I. Samajdar:** Conceptualization, Methodology, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to thank Indian Space Research Organization (ISRO) for financial support and supply of the material. We would like to convey our gratitude to Dr. René de Kloe from EDAX™ for his contributions with EBSD structure files and Hough space optimization. The National Facility of Texture and OIM is acknowledged for the EBSD and XRD measurements.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jallcom.2020.157427>.

Funding

This research was supported by Indian Space Research Organization (ISRO), India under Grant (RD/0118-ISROCO0-008).

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